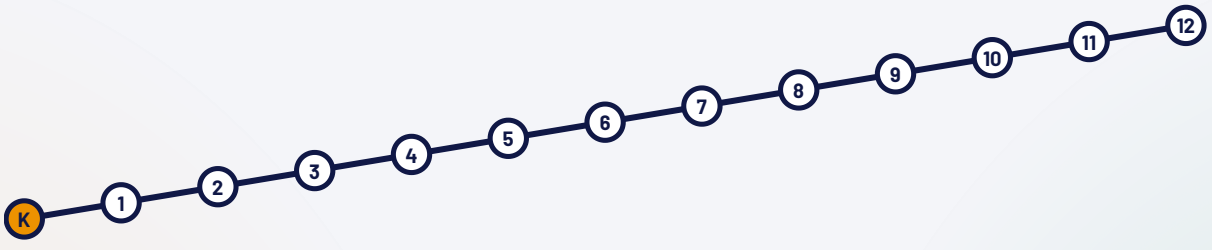
Awsaj Academy

K-12 VERTICAL ALIGNMENT · AN ANALYTICAL WALKTHROUGH

Our Curriculum Journey: Math

What our teachers' ratings tell us about how our standards are taught, from Kindergarten to Grade 12, with questions and options for our part in the story.



These findings come from the ratings you entered as teachers. They show what teachers reported as taught across each grade (2,709 ratings, completed November 2026), not what students know or can do, and they do not judge any teacher. Every item links to the data on the Vertical Alignment site.

Contents

1	**Math, K-12** Math from Kindergarten to Grade 12.	**3**
2	**About the data (at the back)** Where the data comes from, how to read the ratings, the words we use, and the limits.	**18**

Each part follows the same path: the picture (charts), what the numbers say, what we notice, questions for us, and our part in the story (options for improving the curriculum, working across grade levels, knowing the bands above and below, and a concrete action). The data behind each item is shown in brackets, and each item links to the matching page of the Vertical Alignment site. About the data (where the data comes from, how to read the ratings, the words we use and the limits) is at the back. 21 pages, A4. The bookmarks jump to each part.

Taught in depth Introduced / exposed Not taught

MATH, KINDERGARTEN TO GRADE 12

Math, K-12

Across K-12, teachers reported that 62% of the 460 Math ratings (406 standards) are at least introduced, and 17% are taught in depth. Math has real peaks: Grade 2 (62% in depth), place value in K-5 (60%) and Grade 9 equations (60%).

17%
taught in depth

62%
at least introduced

38%
not taught

460
grade-level ratings

What you are seeing. Four headline figures for this part: the share taught in depth, at least introduced and not taught, and how many ratings they come from.

What it adds to our story. This is the starting picture for this part. The slides that follow show where these numbers come from and what they might mean for us.

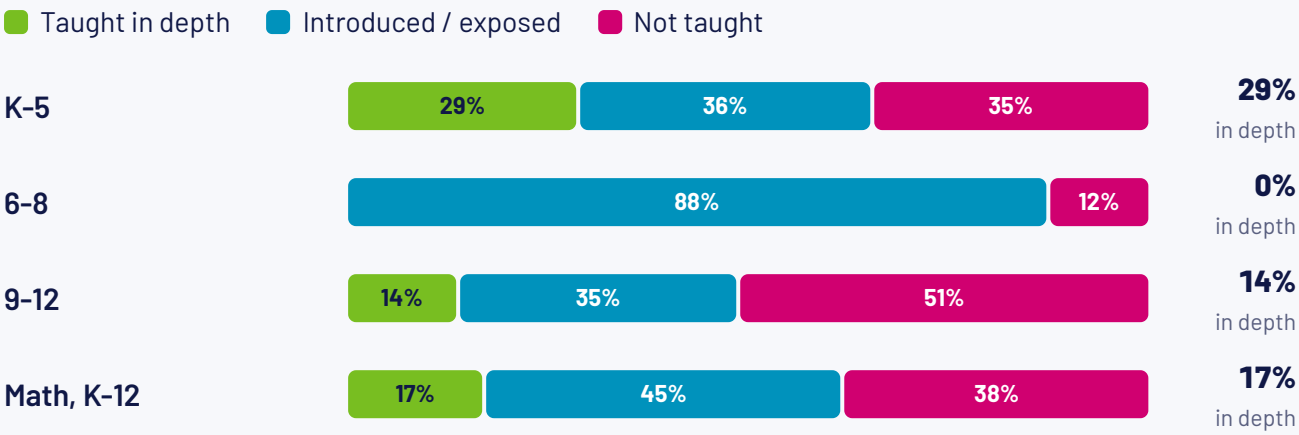
These findings come from the ratings you entered as teachers. They show what teachers reported as taught across each grade, not what students know or can do. We are sharing them back so we can see the whole picture together and decide how we move forward. Each part ends with options for our part in the story. Every item links to the data on the Vertical Alignment site.

THE PICTURE

At a glance

What you are seeing. Each bar shows all the ratings for one subject: green is taught in depth, blue is introduced, pink is not taught.

What it adds to our story. Math introduces most of its standards. Depth is strongest in K-5 and in Grade 9.



How to read this: each bar is 100% of the grade-level ratings. Green is taught in depth, blue is introduced or exposed, pink is not taught. Essential Elements are counted apart.

Open the coverage map on the site ↗

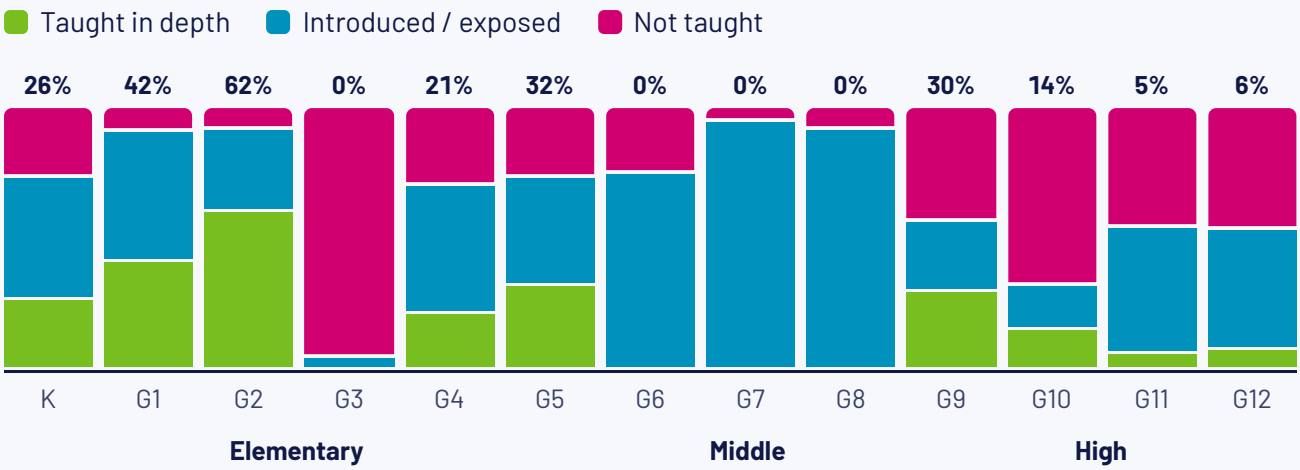
	Rated	In depth	Introduced	Not taught
K-5	**173**	**29%**	**36%**	**35%**
6-8	**81**	**0%**	**88%**	**12%**
9-12	**206**	**14%**	**35%**	**51%**
K-12	**460**	**17%**	**45%**	**38%**

THE PICTURE

Grade by grade

What you are seeing. One column for each grade. The number on top is the share of that grade's standards taught in depth.

What it adds to our story. Strong early grades and a strong Grade 9 frame the middle years, where Math is introduced rather than taken to depth.



How to read this: one column per grade. The number on top is the share taught in depth.

Strand	K	G1	G2	G3	G4	G5	G6	G7	G8	G9	G10	G11	G12
Counting and Cardinality	25%												
Geometry	50%	33%	33%	0%	100%	25%	0%	0%	0%				
Measurement and Data	33%	50%	20%	0%	29%	38%							
Number and Operations in Base Ten	0%	63%	100%	0%	17%	75%							
Operations and Algebraic Thinking	0%	22%	100%	0%	17%	33%							
Number and Operations - Fractions				0%	0%	0%							
Ratios and Proportional Relationships							0%	0%					
Statistics and Probability							0%	0%	0%				
The Expressions and Equations							0%	0%	0%				
The Number System							0%	0%	0%				
Functions									0%				
Arithmetic with Polynomials and Rational Expressions										0%		0%	50%
Building Functions										0%		0%	0%
Creating Equations										50%	100%	0%	0%
Interpreting Categorical and Quantitative Data										11%			
Interpreting Functions										33%		33%	13%
Linear, Quadratic, and Exponential Models										0%		0%	0%
Quantities										0%	100%		
Reasoning with Equations and Inequalities										60%	100%	17%	0%
Seeing Structure in Expressions										0%		0%	50%
The Real Number System										100%			
Circles											0%		
Conditional Probability and the Rules of Probability											0%		0%
Congruence											31%		
Expressing Geometric Properties											0%		0%
Geometric Measurement and Dimension											0%		

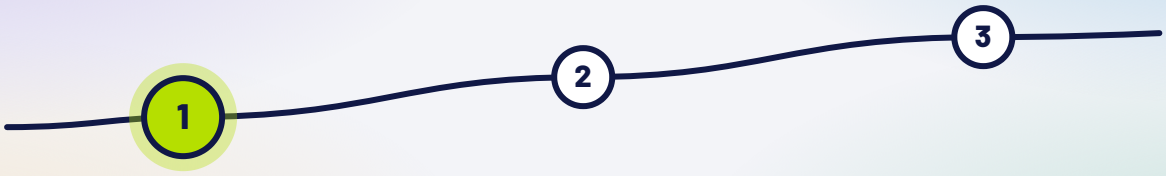
Strand	K	G1	G2	G3	G4	G5	G6	G7	G8	G9	G10	G11	G12
Similarity, Right Triangles, and Trigonometry											0%		0%
Making Inferences and Justifying Conclusions												0%	0%
The Complex Number System												0%	11%
Trigonometric Functions												0%	0%
Using Probability to Make Decisions												0%	
Vector and Matrix Quantities													0%

How to read this: each square is the share of that strand's rated standards taught in depth in that grade. Darker green means more in depth. A blank square means the grade has no standards in that strand.

Open the grade dashboards ↗ Gaps and repetition ↗ Progressions ↗

MATH · OUR CURRICULUM JOURNEY

Section 1: Facts



1 **Facts** YOU ARE HERE

2 Patterns

3 Moving forward

SECTION 1 OF 3

Facts

What you are seeing. The main facts from the ratings, in plain numbers. Each one links to the data on the Vertical Alignment site.

What it adds to our story. Number is our strength. Fractions, ratios and equations, the path to algebra, are the thread to follow.

Bright spots

Where teaching is strongest.

1. **Grade 4 Geometry and Grade 2 Operations are fully in depth.** Both are at 100%. Coverage map ↗
2. **Place value is the strongest strand.** Number and Operations in Base Ten is 60% in depth in K-5: 100% in Grade 2, 75% in Grade 5, 63% in Grade 1. Coverage map ↗
3. **Grade 9 equations.** Reasoning with Equations and Inequalities 60% in depth, Creating Equations 50%, The Real Number System 100%. Grade 9 dashboard ↗
4. **K-2 climbs steadily.** 26%, 42%, 62%: the clearest upward run in any subject. Coverage map ↗

Gaps

Standards no grade teaches, or not taught where they were placed.

5. **Fractions.** 0% in depth across Grades 3 to 5. Not taught: 100% in Grade 3, 42% in Grade 4, 64% in Grade 5. Coverage map ↗
6. **Progressions.** 10 of 18 Math progressions break down on an official Coherence Map link (a published map of which Math standards build on which). One, ratios and proportional relationships, is taught at every step. Progressions ↗
7. **Hand-overs.** Three Grade 9 standards taught in depth build on a Grade 6 or Grade 8 standard rated not taught. All 3 rest on official links. Progressions ↗

8. **Standards no grade teaches.** 70 confirmed: 60 in K-5 and 10 in 6-8, none in high school. Most are in Fractions (19), Operations and Algebraic Thinking (15) and Measurement and Data (15). Gaps and repetition ↗

9. **Not taught where placed.** 77 high school standards, in courses where they were placed by suggestion. Gaps and repetition ↗

Breadth without depth

Standards introduced widely but rarely taken to depth.

10. **Introduced, never in depth.** 186 standards are introduced and never in depth: 63 in K-5, 71 in 6-8, 52 in 9-12. Gaps and repetition ↗

Anomalies

Results that stand out and are worth a second look.

11. **Middle school Math.** 71 of 81 ratings are introduced, and none is yet rated in depth. Coverage map ↗

Other notable

Other facts worth knowing.

12. **Depth by grade.** K 26%, G1 42%, G2 62%, G3 0%, G4 21%, G5 32%, G6 to G8 0%, G9 30%, G10 14%, G11 5%, G12 6%. Coverage map ↗

13. **Priority and power standards.** 21% in depth, 61% introduced, 18% not taught. Grade 9 priorities are 75% in depth and Grade 2 power standards 64%. Priority and power (Audit) ↗

14. **Essential Elements.** 47% in depth K-12: 82% in K-5, 7% in 6-8, 24% in 9-12. Coverage map ↗

15. **Number of standards.** Grades K-8 rate 23 to 34 Math standards each. Grades 9 to 12 rate 43 to 64 each. Coverage map ↗



SECTION 2 OF 3

Patterns

What you are seeing. What we notice when we put the facts side by side: rises, dips and comparisons. These are possibilities to talk about, not conclusions.

What it adds to our story. Following one idea, such as fractions to equations, across the grades shows where students may need more depth.

Trends across the grades

How things rise, fall or alternate from grade to grade.

1. **Math dips in middle school, with strong ground on both sides.** Depth rises through K-2 (26%, 42%, 62%), recovers in Grades 4 and 5 (21%, 32%), rests at "introduced" for three years, and returns in Grade 9 (30%). The middle school years connect two places where we already teach deeply. Grade 9 dashboard ↗

Handoffs

What passes between grades and between schools.

2. **We are strongest in number, thinner in what number leads to.** Counting and place value are solid (place value is 60% in depth). Fractions, ratios, and expressions and equations, the strands that carry students from arithmetic to algebra, are not yet rated in depth from Grade 3 to Grade 8. Coverage map ↗

3. **Fractions are the single clearest thread to follow.** Not in depth in Grades 3 to 5 (0 of 30 ratings), followed by ratios and proportional relationships introduced only in Grades 6 and 7 (6 of 6), followed by Grade 9 equations (60% in depth). Each depends on the one before. Gaps and repetition ↗

4. **Grade 3 is a hinge year.** Multiplication, division, fractions and area all begin in Grade 3. Six progression findings land there, the most of any grade in Math, and 4 of the 10 broken Math progressions trace back to Grade 3 standards rated not taught. Progressions ↗

5. **Kindergarten gaps are foundational ones.** Among the 60 K-5 standards no grade teaches are counting to 100, addition and subtraction word problems, and composing and decomposing numbers within 10. Grade 1 then teaches standards in depth that build on them. Progressions ↗

Essential Elements

Where much teaching may happen at the level students are at.

6. **K-5 Essential Elements tell a second story.** 82% of K-5 Math Essential Elements are in depth (95% in Grades 1 and 5). Much deep Math teaching may be happening at the Essential Element level, which the main ratings do not show. Coverage map ↗

How we rated

Where the ratings may reflect how the scale was read.

7. **In middle school the two sets agree.** Essential Elements are also almost all "introduced" in Grades 6 to 8 (7% in depth). That consistency again points to how "in depth" was read. Coverage map ↗

8. **The dip may be partly how the scale was read.** Middle school Math (71 of 81 ratings) is rated almost uniformly "introduced", and Grade 3 (31 of 32) almost uniformly "not taught". Uniform ratings across a whole grade usually say more about the reading of the scale than about each standard. Grade 3 dashboard ↗

Choices and capacity

What a grade can carry in a year, and what we choose.

9. **High school Math narrows by design or by default.** 51% is not taught, and whole strands (vectors, complex numbers, trigonometric functions, probability for decisions) have little or nothing. Several of these are advanced strands not every pathway includes. Whether that is a choice is worth saying out loud. Coverage map ↗

10. **Priority standards are deeper in high school.** Priority standards are in depth at almost twice the rate of all high school Math standards (27% against 14%): 75% in Grade 9, but 13% and 9% in Grades 11 and 12. Priority and power (Audit) ↗

11. **Capacity shows what a grade carries in Math each year.** A grade takes about 6 Math standards to depth a year, introduces about 16 and does not teach about 14, out of about 35 rated. In K-5 it is about 8 of 29, in middle school none yet of 27 (24 introduced), in high school about 7 of 52. As shares of Math standards: K-5 29% in depth, 36% introduced, 35% not taught; 6-8 0% in depth, 88% introduced, 12% not taught; 9-12 14% in depth, 35% introduced, 51% not taught. Capacity here means how many standards a grade's teaching time actually holds in a year. It is not a target and not a judgement. Coverage map ↗

Toward the SAT

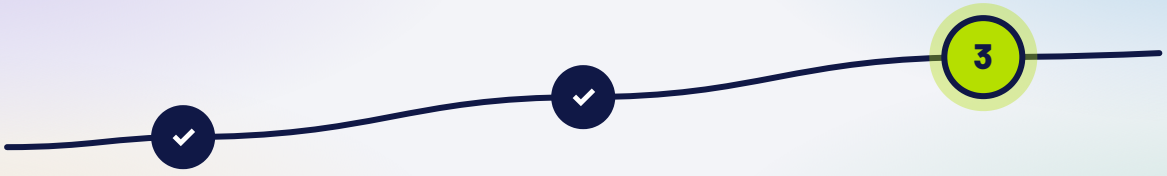
What the ratings suggest about readiness for the SAT.

12. **On the SAT road, algebra is our strongest area.** Algebra is our strongest high school area (about 28% in depth). Data analysis (about 3%) and geometry and trigonometry (about 7%) are thinner, and in middle school all of these are introduced only. Gaps and repetition ↗

Other notable

Other facts worth knowing.

13. **Only 3 standards are in depth in more than one grade.** Creating equations, function notation and solving equations are each in depth in 2 or 3 high school grades. They are also the core of SAT algebra. Coverage map ↗



SECTION 3 OF 3

Moving forward

What you are seeing. Questions for teams to talk about. Each one shows the data that prompted it, and links to where that figure appears on this page.

What it adds to our story. These questions ask how Math teams across grades could agree on a few standards to take deeper together.

Improving the curriculum

What to teach, deepen, place or leave out.

1. **How could each grade take its own data and statistics standards further, given they appear in every band and reach almost no depth after Grade 5?**

   (Data: Measurement and Data 23% in depth in K-5; Statistics and Probability 0% in 6-8; 1 of 38 ratings in high school) see Fact 5

   Grade 5 dashboard ↗

1 **Could middle school pick one strand, such as ratios, and plan it to depth as a trial?**

(Data: All 6 ratio ratings are introduced; none is rated not taught)

Coverage map ↗

2 **Where do Science units already use Math (measurement, data, graphs), and could those lessons take a Math standard to depth?**

(Data: Measurement and Data is 23% in depth in K-5; Statistics and Probability is 0% in 6-8) see Fact 5

Coverage map ↗

3 **Which high school strands have we chosen not to teach, and does each student's pathway still reach what the SAT asks?**

(Data: 51% of high school Math is not taught; 11 of 21 strands have nothing in depth; 77 standards not taught where placed) see Pattern 9

Gaps and repetition ↗

1 **Which 10 Math standards per grade would we protect if we could only take 10 to depth?**

(Data: 17% of Math is in depth K-12; 186 standards are introduced and never in depth) see Fact 10

Gaps and repetition ↗

Working across grade levels

Grades and teams comparing and planning together.

1 **What does Math teaching look like in Grades 1 and 2 that we could learn from across the school?**

(Data: 42% and 62% in depth; Grade 2 place value and operations are both 100%) see Fact 4

Grade 2 dashboard ↗

1 **What would Grade 5 and Grade 6 teachers each want the other to see in their classrooms?**

(Data: Grade 5 is 32% in depth; Grade 6 is 0% in depth and 76% introduced) see Fact 12

Coverage map ↗

Knowing the bands above and below

What the next grade or school relies on.

1 **If we followed one student's path through fractions, ratios and equations from Grade 3 to Grade 9, where would it feel secure and where would it feel thin?**

(Data: Fractions 0% in depth in Grades 3 to 5; ratios and expressions 0% in Grades 6 to 8; Grade 9 equations 60%) see Fact 2

Coverage map ↗

1 **What does Grade 9 algebra need from Grades 6 to 8, and has that list been shared?**

(Data: 3 hand-over gaps on official links, from Grade 6 and Grade 8 equation standards rated not taught)

Progressions ↗

How we rate and plan

How we read the scale and plan our time.

1 **What evidence would show a Math standard has moved from "introduced" to "in depth" in every class: a shared quiz, a common task, reteaching time?**

(Data: 45% of Math ratings are introduced, the most common rating) see At a glance chart

Coverage map ↗

1 **If K-5 Essential Elements are taught in depth, how do we build a bridge from each one to its grade-level standard?**

(Data: 82% of K-5 Math Essential Elements in depth against 29% of the main standards) see Fact 14

Coverage map ↗

2 **What did each team mean by its ratings, and would Grade 3 and middle school rate the same way after a conversation with Grades 2 and 9?**

(Data: Grade 3 rated 31 of 32 not taught; middle school rated 71 of 81 introduced; Grade 2 is 62% and Grade 9 is 30% in depth) see Pattern 8

Coverage map ↗

3 **Which Kindergarten number standards are taught in practice but were rated against a stricter bar?**

(Data: Kindergarten Math is 26% in depth and 26% not taught; counting to 100 and word problems are rated not taught) see Fact 4

Coverage map ↗

1. **If a grade takes about 6 to 8 Math standards to depth a year, which ones carry students furthest toward algebra?**

   (Data: Per grade: about 6 in depth, 16 introduced, 14 not taught, out of 35; K-5 8 of 29, 6-8 0 of 27, 9-12 7 of 52) see Pattern 11

   Coverage map ↗

2. **Is this the Math mix we would choose, and which standards would we move from introduced to in depth, or leave out to make room?**

   (Data: Math mix: 17% in depth, 45% introduced, 38% not taught) see At a glance chart

   Coverage map ↗

Our part in the story

What you are seeing. Options for what we could do, under four headings, each with the data behind it.

What it adds to our story. This is where findings turn into action: small, shared steps we could take together.

Options to discuss and act on. Pick what fits your team, change it, or add your own.

Improving the curriculum

We could agree a short list of Math standards per grade that every class takes to depth, protected in the timetable.

(Data: 17% of Math ratings are in depth; 186 standards are introduced and never in depth) see Fact 10

Gaps and repetition ↗

Working across grade levels

We might hold one moderation session for Grades 2, 3, 5, 6 and 9, each bringing a piece of student work on the same idea.

(Data: Grade 2 62%, Grade 3 0%, Grade 5 32%, Grades 6 to 8 0%, Grade 9 30% in depth) see Fact 12

Coverage map ↗

Knowing the bands above and below

We could follow fractions, ratios and equations as one thread from Grade 3 to Grade 9, with each grade naming its part.

(Data: Fractions 0% in Grades 3 to 5; ratios and expressions 0% in Grades 6 to 8; Grade 9 equations 60%) see Fact 2

Coverage map ↗

A concrete action we could take

We might use the K-5 Essential Elements, where much deep teaching already happens, as the first step of each grade-level plan.

(Data: 82% of K-5 Math Essential Elements in depth against 29% of the main standards) see Fact 14

Coverage map ↗

WHERE THE NUMBERS COME FROM AND HOW TO READ THEM

About the data

This paper walks through what our teachers reported about how our standards are taught, from Kindergarten to Grade 12. It has eight parts: the whole school, the three grade bands, and the four subjects. Each part has three sections: Facts, Patterns, and Moving forward (questions for teams to take forward, and options for our part in the story).

> **These findings come from the ratings you entered as teachers.** They show what teachers reported as taught across each grade, not what students know or can do. We are sharing them back so we can see the whole picture together and decide how we move forward. Each part ends with options for our part in the story. Every item links to the data on the Vertical Alignment site.

ABOUT THE DATA

How to read this paper

> **What you are seeing.** How the ratings were made, what the three ratings mean, and the words we use.
>
> **What it adds to our story.** It helps us read every number fairly: these are teachers' reports of teaching, not measures of what students know.

This paper walks through what our teachers reported about how our standards are taught, from Kindergarten to Grade 12. It has eight parts: the whole school, the three grade bands, and the four subjects. Each part has three sections: Facts, Patterns, and Moving forward (questions for teams to take forward, and options for our part in the story).

Where the numbers come from. Every figure is computed from the ratings sheet of 9 November 2026 (2,166 standards, 2,709 ratings, last rating 8 November) and checked against the Vertical Alignment reports and findings decks.

The three ratings.

Rating	What it means
Taught in depth	ALL classes in the grade consistently teach, reteach, review and assess the standard
Introduced / exposed	Anything in between: some classes, or taught briefly (for example, 3 of 5 classes)

Rating	What it means
Not taught	No class in the grade teaches it

Four things to hold in mind while reading.

- The ratings are what teachers reported as taught across a whole grade. They are not a measure of what students know or can do, and they do not judge any teacher.
- "In depth" is all or nothing. Teams may have read that more or less strictly, so a low figure can reflect a stricter reading, not less teaching. Where this matters most, the paper says so. Where a team rated nothing in depth, we read it as careful use of an all-or-nothing scale, not as an absence of teaching.
- Each team rated its own subject's standards. Much literacy (vocabulary, academic language, speaking, writing, reading non-fiction) is also taught through Science and Social Studies, and that teaching may not show in the ELA ratings.
- Shares are of the main grade-level ratings (one standard in one grade). Essential Elements (the alternate standards) are counted apart.

We cannot teach everything equally. No school can take every standard to depth, and that is not the goal. The goal is to choose on purpose: which standards we teach in depth, which we introduce, and which we leave out, and to plan that proportion together across the grades. The ratings show the mix we have now, so we can decide the mix we want.

Instructional capacity. How many standards a grade actually takes to depth, introduces or leaves out in a year, worked out from the ratings (the average for one grade). It is not a target and not a judgement: it describes what our teaching time holds now, so we can decide together how to use it.

Categories. In every part, each section groups its items under the same categories, in the same order, so a team can follow one kind of observation across the whole school, its band and its subject. Facts: Bright spots, Gaps, Breadth without depth, Anomalies and Other notable. Patterns: Trends across the grades, Handoffs, Language and literacy, Essential Elements, How we rated, Choices and capacity, Toward the SAT and Other notable. Moving forward: Improving the curriculum, Working across grade levels, Knowing the bands above and below, and How we rate and plan. A category with nothing in it is left out.

How each part is laid out. Three sections, each opening with a small map that shows where you are: Facts, then Patterns, then Moving forward (questions to ponder, then options for our part in the story: improving the curriculum, working across grade levels, knowing the bands above and below, and a concrete action). Every pattern carries its figures in the text, and every question ends with the data behind it, in brackets and italics. The questions are invitations, not conclusions. Our strengths are named alongside the places where we might build.

ABOUT THE DATA

Method, sources and limits

> **What you are seeing.** How the ratings were made, what the three ratings mean, and the words we use.
>
> **What it adds to our story.** It helps us read every number fairly: these are teachers' reports of teaching, not measures of what students know.

Every figure in this paper was computed from the ratings sheet and cross-checked against the site. Where the site's band and K-12 views differ (progressions), the paper says which it uses. Percentages are rounded half up, as on the site.

What was read

Source	How it was used	How fully it was read
Ratings sheet, "All Standards and Ratings K-12" (9 Nov 2026)	Every figure	All 2,166 rows
Organized ratings workbook	Cross-check of totals	All tabs
20 Findings decks (band and subject)	Cross-check, questions and next-step themes	All slides and speaker notes
K-12 Reports PDF (118 pages)	Cross-check	First 80 pages scanned by search; a few read directly
7 other Reports PDFs	Cross-check of headline figures	Scanned by search, not read page by page

How terms are used

- **Rated standards.** One rating is one standard in one grade. A standard shared by several grades is counted once for each grade that rated it.
- **At least introduced.** Rated in depth or introduced.
- **No grade teaches it (confirmed).** Every grade that could teach the standard rated it not taught.
- **Not taught where placed.** The grade it was placed in rated it not taught, and other grades that could teach it were not asked.
- **Introduced, never in depth (met but not owned).** Introduced in at least one grade and in depth in none.
- **Progression finding.** An automatic check on a chain of linked standards: a step not taught, or a standard in depth while the one it builds on is not taught.
- **Taught at every step.** Every standard in the chain, across two or more grades, is at least introduced.

- **Progression.** A chain of linked standards across grades. Official links are the Coherence Map (Math), anchor standards (ELA) and the standards' own structure (Science, Social Studies). Other links are suggestions.
- **Instructional capacity.** The average number of standards one grade rated in depth, introduced and not taught in a year (all ratings divided by the number of grades), rounded.
- **Hand-over gap.** A standard taught in depth while a standard it builds on, placed in an earlier school, is rated not taught.
- **Priority and power standards.** Standards the school's own documents name: power standards in K-5, priority standards in 6-12.

Limits to keep in view

- Ratings are teachers' judgements of teaching across a grade. They are not observations and not student results.
- Grades may have read "in depth" differently. Uniform ratings across a whole grade are a reason to check how the scale was read; the paper points these out so teams can discuss them.
- The ratings are what teachers reported as taught. They do not measure what students know or can do.
- Each team rated its own subject. Literacy taught in Science and Social Studies, and Math taught in Science, may not show in the ELA and Math ratings.
- Middle school Science and high school standards were placed in grades and courses by suggestion. "Not taught" there can mean "taught in another year". 24 K-5 Social Studies standards are also not taught in the grade where they were placed.
- Band counts of progressions look only at steps inside the band, so they do not add up to the K-12 figure.
- Literacy audit figures come from the 2026 PK-5 literacy audit of weekly plans.
- Some strands have few standards in a grade, so a single rating can move a percentage a long way. Counts are given where this matters.
- Suggested links between standards (most ELA, Science and Social Studies hand-overs) have not been checked by teachers.

Status. This is a draft for the Vertical Alignment Committee. Each finding should be checked with the teams concerned before it is treated as settled. The planned review of weekly and unit plans (Part 2) will test many of these patterns against what is actually planned.